

# Natural Language to SPARQL and SPARQL to Natural Language Student project- KOS course

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# Objective

## Develop a system that can

- Convert natural language prompts into executable SPARQL queries.
- Convert SPARQL queries into natural language
- Analyze feasibility
- Study prompt-engineering strategies
- Review existing libraries and tools

## Resources

- Example data set: [Cédric Viaccoz / rdf-coding-dojo-2023 · GitLab](#). You can choose your own dataset
- Local LLM: Hactar (<https://hactar.unige.ch>) ask for a login account: Mehdi Chouaibi or Quentin Morsh

# Scope and feasibility analysis



## Easily feasible

Ex: mapping simple, structured language phrases such « Show me all movies directed by Spielberg» or generating natural language from simple SELECT queries with clear subject-predicate-object patterns.



## Challenging

Ex: Converting nested SPARQL queries (ex: with OPTIONAL, UNION subqueries) into natural language

# Prompt Engineering Strategies

Ask the model to reason step-by-step

Include ontology or schema hints in the prompt (ex: «Use Dbpedia ontology»)

Provide examples in few-shot prompting to guide the model

Break complex questions into simple sub-questions

# How to: Example

- Build a simple prototype
  - Source example: Dbpedia
- Tech Stack
  - Python, SPARQLEngine, streamlit UI, CLI
- Librairies
  - Rdfliib, transformes, spacy

# Existing libraries and tools

- <https://jena.apache.org/>
- <https://rdflib.readthedocs.io/en/stable/>
- <https://huggingface.co/aksw/text2sparql-S>
- <https://github.com/xiaoyuin/tntspa>
- <https://github.com/huggingface/datasets>
- <https://huggingface.co/models?search=sparql>